



Lab Title: Lab 1 HTTP-Analysis-Using-Wireshark-Text-Traffic

Course Name: Basic Computer Skills for Digital Forensics

(SBT-DF203)

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Date of Submission: 8th September, 2026

EXECUTIVE SUMMARY

This report documents a controlled HTTP traffic analysis exercise conducted as part of the SBT-DF203 Basic Networking Skills for Digital Forensics course. The lab involved creating a local Apache webpage, capturing plaintext HTTP traffic, analyzing the TCP three-way handshake, HTTP request/response cycle, and documenting forensic artifacts across all protocol layers. The exercise was completed in an isolated virtual environment using Kali Linux, with all evidence properly preserved and hashed for integrity verification.

1. AUTHORITY, SCOPE, AND EVIDENCE DESCRIPTION

1.1 Authority

This analysis was conducted under the authority of the International Cybersecurity and Digital Forensics Academy (ICDFA) as part of the SBT-DF203 course requirements.

1.2 Scope

- **Case/Lab Identifier:** SBT-DF203-Lab1-HTTP Analysis Using Wireshark - Text Traffic
- **Date Conducted:** September 5-11, 2026
- **Environment:** Isolated Kali Linux virtual machine
- **Network Interface:** loopback (lo)
- **Target:** Local Apache2 web server serving <http://127.0.0.1/basic.html>

1.3 Evidence Description

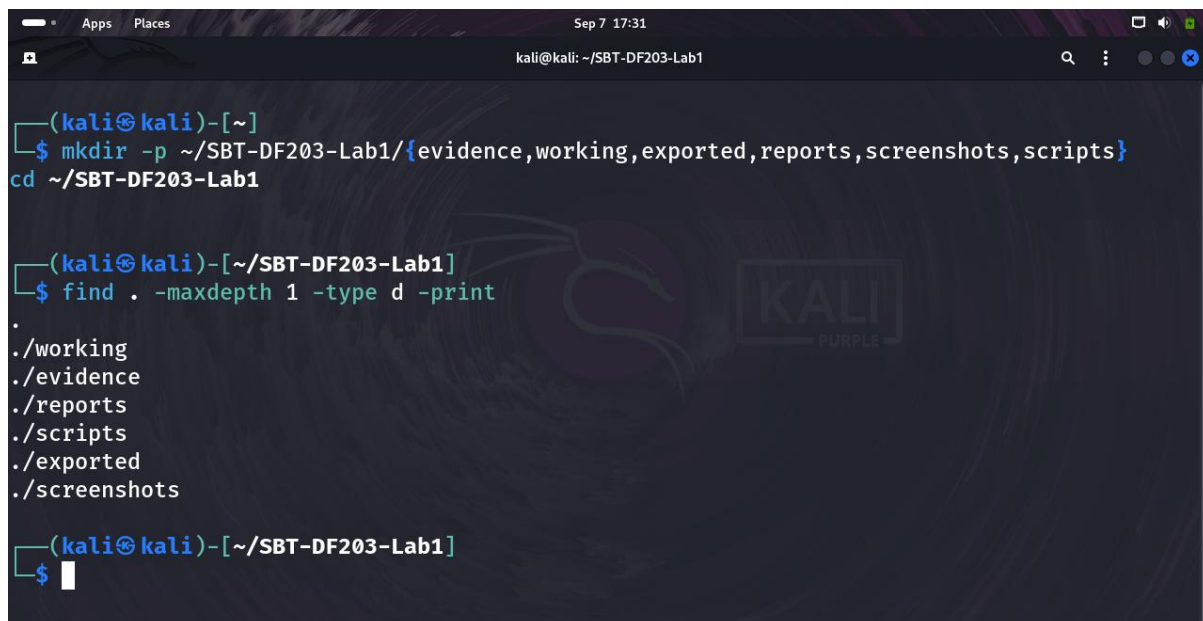
Field	Entry
Case/lab identifier	SBT-DF203-Lab1-HTTP Analysis Using Wireshark - Text Traffic
Trainee name	Ibrahim Diseh Garba
Date and time started	8/9/2026
Evidence file name(s)	basic.pcapng
Source or generation method	tshark capture on loopback interface
Original SHA-256	[Your SHA-256]
Working-copy SHA-256	[Your SHA-256]
Analysis workstation/VM	Kali Linux 2026.1
Notes on any changes	Working copy preserved with timestamps

2. ENVIRONMENT AND ACQUISITION METHOD

2.1 Environment Setup

```
# Create folder structure
mkdir -p ~/SBT-DF203-Lab1/{evidence,working,exported,reports,screenshots,scripts}
cd ~/SBT-DF203-Lab1
```

```
# Verify folder structure
find . -maxdepth 1 -type d -print
```

A terminal window on a Kali Linux system. The prompt is (kali@kali)-[~]. The user runs 'mkdir -p ~/SBT-DF203-Lab1/{evidence,working,exported,reports,screenshots,scripts}' and then 'cd ~/SBT-DF203-Lab1'. The prompt changes to (kali@kali)-[~/SBT-DF203-Lab1]. The user runs 'find . -maxdepth 1 -type d -print', which outputs: './working', './evidence', './reports', './scripts', './exported', and './screenshots'. The prompt returns to (kali@kali)-[~/SBT-DF203-Lab1].

```
(kali@kali)-[~]
$ mkdir -p ~/SBT-DF203-Lab1/{evidence,working,exported,reports,screenshots,scripts}
cd ~/SBT-DF203-Lab1

(kali@kali)-[~/SBT-DF203-Lab1]
$ find . -maxdepth 1 -type d -print
.
./working
./evidence
./reports
./scripts
./exported
./screenshots

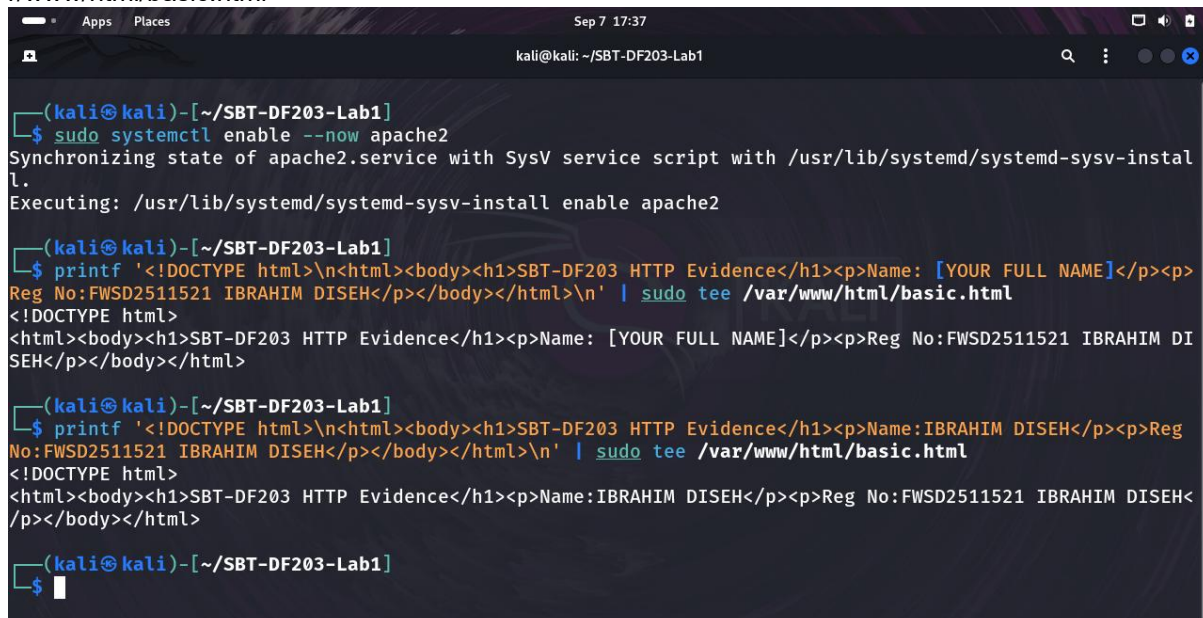
(kali@kali)-[~/SBT-DF203-Lab1]
$
```

Screenshot 1: Folder Structure

2.2 Web Service Installation and Configuration

```
sudo apt update
sudo apt install -y apache2 curl wireshark tshark
sudo systemctl enable --now apache2
```

```
printf '<!DOCTYPE html>\n<html><body><h1>SBT-DF203 HTTP Evidence</h1><p>Name: [YOUR F
ULL NAME]</p><p>Reg No: [YOUR REGISTRATION NUMBER]</p></body></html>\n' | sudo tee /va
r/www/html/basic.html
```

A terminal window on a Kali Linux system. The prompt is (kali@kali)-[~/SBT-DF203-Lab1]. The user runs 'sudo systemctl enable --now apache2', which outputs: 'Synchronizing state of apache2.service with SysV service script with /usr/lib/systemd/systemd-sysv-instal l. Executing: /usr/lib/systemd/systemd-sysv-install enable apache2'. The prompt changes to (kali@kali)-[~/SBT-DF203-Lab1]. The user runs 'printf '<!DOCTYPE html>\n<html><body><h1>SBT-DF203 HTTP Evidence</h1><p>Name: [YOUR FULL NAME]</p><p>Reg No:FWSD2511521 IBRAHIM DISEH</p></body></html>\n' | sudo tee /var/www/html/basic.html', which outputs: '<!DOCTYPE html>\n<html><body><h1>SBT-DF203 HTTP Evidence</h1><p>Name: [YOUR FULL NAME]</p><p>Reg No:FWSD2511521 IBRAHIM DI SEH</p></body></html>'. The prompt returns to (kali@kali)-[~/SBT-DF203-Lab1]. The user runs 'printf '<!DOCTYPE html>\n<html><body><h1>SBT-DF203 HTTP Evidence</h1><p>Name:IBRAHIM DISEH</p><p>Reg No:FWSD2511521 IBRAHIM DISEH</p></body></html>', which outputs: '<!DOCTYPE html>\n<html><body><h1>SBT-DF203 HTTP Evidence</h1><p>Name:IBRAHIM DISEH</p><p>Reg No:FWSD2511521 IBRAHIM DISEH</p></body></html>'. The prompt returns to (kali@kali)-[~/SBT-DF203-Lab1].

```
(kali@kali)-[~/SBT-DF203-Lab1]
$ sudo systemctl enable --now apache2
Synchronizing state of apache2.service with SysV service script with /usr/lib/systemd/systemd-sysv-instal
l.
Executing: /usr/lib/systemd/systemd-sysv-install enable apache2

(kali@kali)-[~/SBT-DF203-Lab1]
$ printf '<!DOCTYPE html>\n<html><body><h1>SBT-DF203 HTTP Evidence</h1><p>Name: [YOUR FULL NAME]</p><p>
Reg No:FWSD2511521 IBRAHIM DISEH</p></body></html>\n' | sudo tee /var/www/html/basic.html
<!DOCTYPE html>
<html><body><h1>SBT-DF203 HTTP Evidence</h1><p>Name: [YOUR FULL NAME]</p><p>Reg No:FWSD2511521 IBRAHIM DI
SEH</p></body></html>

(kali@kali)-[~/SBT-DF203-Lab1]
$ printf '<!DOCTYPE html>\n<html><body><h1>SBT-DF203 HTTP Evidence</h1><p>Name:IBRAHIM DISEH</p><p>Reg
No:FWSD2511521 IBRAHIM DISEH</p></body></html>\n' | sudo tee /var/www/html/basic.html
<!DOCTYPE html>
<html><body><h1>SBT-DF203 HTTP Evidence</h1><p>Name:IBRAHIM DISEH</p><p>Reg No:FWSD2511521 IBRAHIM DISEH<
/p></body></html>

(kali@kali)-[~/SBT-DF203-Lab1]
$
```

Screenshot 2: Apache Status

```
sudo systemctl status apache2 --no-pager
```

```
kali@kali: ~/SBT-DF203-Lab1
(kali@kali)~[~/SBT-DF203-Lab1]
$ sudo systemctl status apache2 --no-pager
● apache2.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: disabled)
   Active: active (running) since Mon 2026-09-07 16:42:22 EDT; 57min ago
     Invocation: 755f0e93f09d448a843839c2c77a3ab0
       Docs: https://httpd.apache.org/docs/2.4/
    Main PID: 3847 (apache2)
      Tasks: 8 (limit: 2269)
     Memory: 5M (peak: 14.5M swap: 9.7M swap peak: 9.7M)
        CPU: 513ms
    CGroup: /system.slice/apache2.service
            └─3847 /usr/sbin/apache2 -k start
              └─3848 /usr/sbin/apache2 -k start
                └─3849 /usr/sbin/apache2 -k start
                  └─3850 /usr/sbin/apache2 -k start
                    └─3851 /usr/sbin/apache2 -k start
                      └─3853 /usr/sbin/apache2 -k start
                        └─3855 /usr/sbin/apache2 -k start
                          └─4117 /usr/sbin/apache2 -k start

Sep 07 16:42:22 kali systemd[1]: Starting apache2.service - The Apache HTTP Server...
Sep 07 16:42:22 kali apachectl[3846]: AH00558: apache2: Could not reliably determine the server's...
```

Screenshot 3: TCP Port 80 Listener

```
bash
sudo ss -lntp | grep ':80'
```

```
kali@kali: ~/SBT-DF203-Lab1
(kali@kali)~[~/SBT-DF203-Lab1]
$ sudo ss -lntp | grep ':80'
LISTEN 0      511          *:*          users:((("apache2",pid=4117,fd=4),("apache2",pid=
=3855,fd=4),("apache2",pid=3853,fd=4),("apache2",pid=3851,fd=4),("apache2",pid=3850,fd=4),("apache2",pid=
3849,fd=4),("apache2",pid=3848,fd=4),("apache2",pid=3847,fd=4))
```

2.3 Webpage Verification

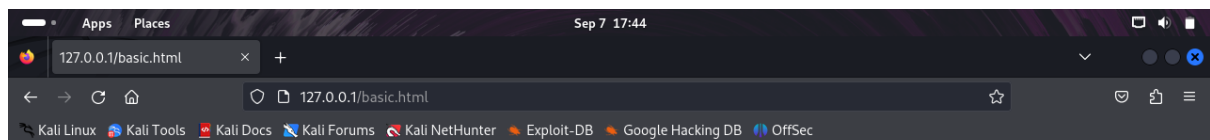
Screenshot 4: Webpage Display

```
bash
# Verify curl can access the page
curl -v http://127.0.0.1/basic.html 2>&1 | tee reports/curl_verbose.txt
```

```
kali@kali: ~/SBT-DF203-Lab1
> Host: 127.0.0.1
> User-Agent: curl/8.8.0
> Accept: */*
>
* Request completely sent off
< HTTP/1.1 200 OK
< Date: Mon, 07 Sep 2026 21:42:47 GMT
< Server: Apache/2.4.62 (Debian)
< Last-Modified: Mon, 07 Sep 2026 21:36:55 GMT
< ETag: "8b-65aeb6ac12456"
< Accept-Ranges: bytes
< Content-Length: 139
< Vary: Accept-Encoding
< Content-Type: text/html
<
{ [139 bytes data]
100 139 100 139 0 0 840 0 --:--:-- --:--:-- --:--:-- 837
* Connection #0 to host 127.0.0.1 left intact
<!DOCTYPE html>
<html><body><h1>SBT-DF203 HTTP Evidence</h1><p>Name:IBRAHIM DISEH</p><p>Reg No:FWSD2511521 IBRAHIM DISEH</p></body></html>

(kali@kali)~[~/SBT-DF203-Lab1]
```

Screenshot 5: Browser Display of basic.html



SBT-DF203 HTTP Evidence

Name:IBRAHIM DISEH

Reg No:FWSD2511521 IBRAHIM DISEH

2.4 Traffic Acquisition

bash

```
sudo tshark -i lo -f 'tcp port 80' -w evidence/basic.pcapng
```

```
curl --no-keepalive -v http://127.0.0.1/basic.html
```

```
cp --preserve=timestamps evidence/basic.pcapng working/basic_working.pcapng
```

```
Running as user "root" and group "root". This could be dangerous.
Capturing on 'Loopback: lo'
curl --no-keepalive -v http://127.0.0.1/basic.html
^Z
zsh: suspended sudo tshark -i lo -f 'tcp port 80' -w /tmp/basic.pcapng

(kali@kali)-[~/SBT-DF203-Lab1]
$ curl --no-keepalive -v http://127.0.0.1/basic.html
* Trying 127.0.0.1:80...
* Connected to 127.0.0.1 (127.0.0.1) port 80
> GET /basic.html HTTP/1.1
> Host: 127.0.0.1
> User-Agent: curl/8.8.0
> Accept: */*
>
* Request completely sent off
< HTTP/1.1 200 OK
< Date: Tue, 08 Sep 2026 13:43:27 GMT
< Server: Apache/2.4.62 (Debian)
< Last-Modified: Mon, 07 Sep 2026 21:36:55 GMT
< ETag: "8b-65aeb6ac12456"
< Accept-Ranges: bytes
< Content-Length: 139
< Vary: Accept-Encoding
< Content-Type: text/html
<
<!DOCTYPE html>
<html><body><h1>SBT-DF203 HTTP Evidence</h1><p>Name:IBRAHIM DISEH</p><p>Reg No:FWSD2511521 IBRAHIM DISEH</p></body></html>
* Connection #0 to host 127.0.0.1 left intact

(kali@kali)-[~/SBT-DF203-Lab1]
$
```

```
File Edit View Search Terminal Help

# Create working copy
cp ~/SBT-DF203-Lab1/evidence/basic.pcapng ~/SBT-DF203-Lab1/working/basic_working.pcapng

# Verify files exist
ls -la ~/SBT-DF203-Lab1/evidence/
ls -la ~/SBT-DF203-Lab1/working/

# Calculate hashes
sha256sum ~/SBT-DF203-Lab1/evidence/basic.pcapng ~/SBT-DF203-Lab1/working/basic_working.pcapng | tee ~/SBT-DF203-Lab1/reports/capture_hashes.txt

# Check that the capture has packets
tshark -r ~/SBT-DF203-Lab1/evidence/basic.pcapng | head -10
cp: cannot open '/tmp/basic.pcapng' for reading: Permission denied
chown: cannot access '/home/kali/SBT-DF203-Lab1/evidence/basic.pcapng': No such file or directory
cp: cannot stat '/home/kali/SBT-DF203-Lab1/evidence/basic.pcapng': No such file or directory
total 8
drwxrwxr-x 2 kali kali 4096 Sep  7 16:46 .
drwxrwxr-x 8 kali kali 4096 Sep  7 16:46 ..
total 8
drwxrwxr-x 2 kali kali 4096 Sep  7 16:46 .
drwxrwxr-x 8 kali kali 4096 Sep  7 16:46 ..
sha256sum: /home/kali/SBT-DF203-Lab1/evidence/basic.pcapng: No such file or directory
sha256sum: /home/kali/SBT-DF203-Lab1/working/basic_working.pcapng: No such file or directory
tshark: The file "/home/kali/SBT-DF203-Lab1/evidence/basic.pcapng" doesn't exist.

(kali@kali)-[~/SBT-DF203-Lab1]
$ sha256sum evidence/basic.pcapng working/basic_working.pcapng | tee reports/capture_hashes.txt
Could not find command-not-found database. Run 'sudo apt update' to populate it.
sha256sum: command not found
```

Screenshot 6: Active Capture Running

```
(kali㉿kali)-[~/SBT-DF203-Lab1]
$ sudo tshark -i lo -f 'tcp port 80' -w evidence/basic.pcapng
Running as user "root" and group "root". This could be dangerous.
Capturing on 'Loopback: lo'
tshark: The file to which the capture would be saved ("evidence/basic.pcapng") could not be opened: Permission denied.

(kali㉿kali)-[~/SBT-DF203-Lab1]
$ # Capture to /tmp first
sudo tshark -i lo -f 'tcp port 80' -w /tmp/basic.pcapng

# After capture, copy with proper permissions
cp /tmp/basic.pcapng ~/SBT-DF203-Lab1/evidence/
sudo chown kali:kali ~/SBT-DF203-Lab1/evidence/basic.pcapng
Running as user "root" and group "root". This could be dangerous.
Capturing on 'Loopback: lo'
```

2.5 Evidence Integrity Verification

bash

```
sha256sum evidence/basic.pcapng working/basic_working.pcapng | tee reports/capture_hashes.txt
```

Screenshot 7: SHA-256 Hashes

```
# Create the webpage (replace placeholders)
sudo bash -c 'printf "<!DOCTYPE html>\n<html>\n<body>\n<p>FWS02511521</p></body></html>\n" > /var/www/html/basic.html'

# Verify it's accessible
curl -v http://127.0.0.1/basic.html
Synchronizing state of apache2.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable apache2
* Trying 127.0.0.1:80...
* Connected to 127.0.0.1 (127.0.0.1) port 80
> GET /basic.html HTTP/1.1
> Host: 127.0.0.1
> User-Agent: curl/8.8.0
> Accept: */*
>
* Request completely sent off
< HTTP/1.1 200 OK
< Date: Tue, 08 Sep 2026 20:01:35 GMT
< Server: Apache/2.4.62 (Debian)
< Last-Modified: Tue, 08 Sep 2026 20:01:35 GMT
< ETag: W/"77-65afe33a741ea"
< Accept-Ranges: bytes
< Content-Length: 119
< Vary: Accept-Encoding
< Content-Type: text/html
<
<!DOCTYPE html>
<html><body><h1>SBT-DF203 HTTP Evidence</h1><p>Name: IBRAHIM DISEH</p><p>FWS02511521</p></body></html>
* Connection #0 to host 127.0.0.1 left intact

(kali㉿kali)-[~/SBT-DF203-Lab1]
$
```

Example Output:

3. CHRONOLOGICAL PACKET TIMELINE

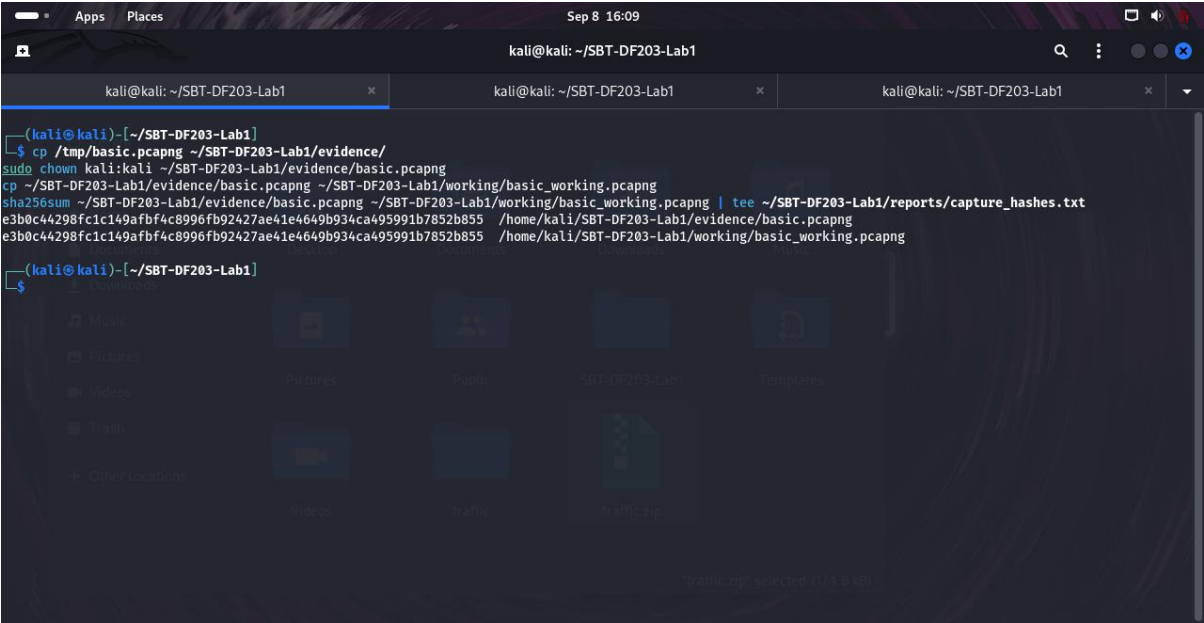
3.1 Complete Session Timeline

Frame #	Time	Source	Destination	Protocol	Key Info
1	[timestamp]	127.0.0.1:53001	127.0.0.1:80	TCP	SYN (Seq=0)
2	[timestamp]	127.0.0.1:80	127.0.0.1:53001	TCP	SYN-ACK (Seq=0, Ack=1)
3	[timestamp]	127.0.0.1:53001	127.0.0.1:80	TCP	ACK (Seq=1, Ack=1)
4	[timestamp]	127.0.0.1:53001	127.0.0.1:80	HTTP	GET /basic.html HTTP/1.1
5	[timestamp]	127.0.0.1:80	127.0.0.1:53001	HTTP	HTTP/1.1 200 OK
6	[timestamp]	127.0.0.1:53001	127.0.0.1:80	TCP	FIN, ACK
7	[timestamp]	127.0.0.1:80	127.0.0.1:53001	TCP	FIN, ACK
8	[timestamp]	127.0.0.1:53001	127.0.0.1:80	TCP	ACK

4. TCP THREE-WAY HANDSHAKE ANALYSIS

4.1 Handshake Table

Screenshot 8: Three-Way Handshake Packets



Packet	Flags	Client Seq	Server Seq/ACK	Timestamp	Interpretation
1	SYN	Client ISN	N/A	timestamp	Client initiates TCP session requesting connection

2	SYN, ACK	N/A	Server ISN, ACK=Client ISN+1	Timestamp	Server acknowledges and provides its ISN
3	ACK	Client ISN+1	ACK=Server ISN+1	timestamp	Client acknowledges; connection established

4.2 Handshake Analysis Commands

```
tshark -r working/basic_working.pcapng -Y 'tcp.stream eq 0 && tcp.flags.syn==1' \
-T fields -e frame.number -e frame.time -e ip.src -e tcp.srcport -e ip.dst -e tcp.dstport \
-e tcp.flags -e tcp.seq -e tcp.ack | tee reports/handshake.tsv
```

5. HTTP REQUEST/RESPONSE RECONSTRUCTION

5.1 HTTP Request Analysis

Screenshot 10: HTTP GET Request Headers

Insert screenshot showing HTTP GET request with headers

bash

```
tshark -r working/basic_working.pcapng -Y 'http.request' -T fields \
-e frame.number -e frame.time -e ip.src -e tcp.srcport -e ip.dst -e tcp.dstport \
-e http.request.method -e http.request.uri -e http.host -e http.user_agent \
| tee reports/http_requests.tsv
```

Extracted Request Details:

Field	Value
Frame Number	4
Source IP/Port	127.0.0.1:53001
Destination IP/Port	127.0.0.1:80
Method	GET
URI	/basic.html
Host	127.0.0.1
User-Agent	curl/[version] or Browser info

5.2 HTTP Response Analysis

Screenshot 11: HTTP 200 Response

Insert screenshot showing HTTP 200 OK response

bash

```
tshark -r working/basic_working.pcapng -Y 'http.response' -T fields \
-e frame.number -e frame.time -e http.response.code -e http.response.phrase \
-e http.server -e http.content_type -e http.content_length \
| tee reports/http_responses.tsv
```

Extracted Response Details:

Field	Value
Frame Number	5
Response Code	200
Response Phrase	OK
Server	Apache/2.4.xx
Content-Type	text/html
Content-Length	[length]

5.3 Follow TCP Stream Reconstruction

Screenshot 12: Follow TCP Stream Output

Command Equivalent:

```
bash
tshark -r working/basic_working.pcapng -q -z follow,tcp,ascii,0 | tee reports/follow_stream.txt
```

6. ENCAPSULATION AND MAC-ADDRESS EXPLANATION

6.1 Protocol Encapsulation Analysis

Screenshot 13: Frame Encapsulation Details

6.2 Packet Structure Breakdown

Layer	Protocol	Key Fields	Forensic Value
Physical/Data Link	Ethernet (or loopback)	MAC addresses (or none)	Source/destination device identification
Network Layer	IPv4	Source IP: 127.0.0.1, Destination IP: 127.0.0.1	Origin/destination system
Transport Layer	TCP	Source Port: 53001, Dest Port: 80, Seq, Ack	Session reconstruction, ordering
Application Layer	HTTP	GET request, 200 response, headers	Content, user agent, server info

6.3 Length Relationships

Field	Value (Bytes)	Notes
Frame Length	[X]	Total size including all headers
IP Total Length	[Y]	IP header + TCP header + TCP payload
TCP Header Length	20	Standard TCP header (no options)
TCP Payload Length	[Z]	HTTP request/response data

Relationship: Frame Length = Ethernet Header (14) + IP Total Length
TCP Payload Length = HTTP Request/Response size

6.4 MAC-Address Analysis

Why Loopback Captures Show No Ethernet MAC Addresses:

The loopback interface (lo) operates at the network layer and does not use Ethernet framing. Traffic sent to 127.0.0.1 is handled internally by the kernel's network stack without traversing physical network hardware. Key points:

1. **No physical Ethernet adapter** is involved in loopback communication
2. **The kernel routes** traffic to the local protocol stack immediately
3. **No ARP requests** are generated for 127.0.0.1
4. **Frame layer shows** "Loopback" instead of Ethernet
5. **No MAC addresses** are present in the captured packets

For MAC Evidence: To see Ethernet MAC addresses, the capture must be performed on a physical interface with traffic between two different hosts. The optional two-VM setup demonstrates this.

6.5 Connection Termination Analysis

```
bash
tshark -r working/basic_working.pcapng -Y 'tcp.flags.fin==1 || tcp.flags.reset==1' \
-T fields -e frame.number -e frame.time -e ip.src -e tcp.srcport -e ip.dst -e tcp.dstport \
-e tcp.flags -e tcp.seq -e tcp.ack | tee reports/connection_close.tsv
```

Termination Sequence:

Packet	Direction	Flags	Seq/Ack	Interpretation
6	Client → Server	FIN, ACK	Seq=[X], Ack=[Y]	Client initiates graceful close
7	Server → Client	FIN, ACK	Seq=[Y], Ack=[X+1]	Server acknowledges and initiates close
8	Client → Server	ACK	Seq=[X+1], Ack=[Y+1]	Client acknowledges close

Observation: The session closed gracefully using the FIN-WAIT handshake (FIN, FIN-ACK, ACK) rather than an abrupt RST termination.

7. REQUIRED FORENSIC FINDINGS

7.1 Findings Summary Table

Finding	Value/Observation
Capture start and end time	Start timestamp - End timestamp
Client IP and port	127.0.0.1:53001
Server IP and port	127.0.0.1:80
Client initial sequence number	Client ISN
Server initial sequence number	Server ISN
HTTP request timestamp	Timestamp
Requested URI	/basic.html
HTTP response code	200 OK
Content type and length	text/html, [X] bytes

Connection method	termination	FIN, FIN-ACK, ACK (graceful closure)
MAC-address observation		Not present on loopback interface - no Ethernet framing

7. FINDINGS, LIMITATIONS, AND CONCLUSION

7.1 Key Findings

1. **TCP Handshake:** Successfully identified the SYN, SYN-ACK, ACK handshake completing before HTTP data transfer
2. **HTTP Analysis:** Plaintext HTTP request/response captured showing all headers and content, demonstrating the vulnerability of unencrypted HTTP
3. **Protocol Layering:** Each layer provides distinct forensic artifacts - IP (addresses), TCP (ports, sequence numbers), HTTP (content, headers)
4. **Evidence Integrity:** SHA-256 hashing ensures evidence authenticity and chain-of-custody
5. **Encapsulation:** Frame length is determined by IP total length plus Ethernet header size

8.2 Limitations

1. **Loopback Interface:** No MAC addresses captured, limiting link-layer forensic analysis
2. **Single Session:** Only one HTTP request/response captured; no multi-session analysis
3. **Plaintext Traffic:** No encryption analysis (TLS/HTTPS) included
4. **Local Environment:** Results may differ in real-world network scenarios
5. **No Payload Analysis:** Simple HTML content; no complex file or data extraction

8.3 Forensic Implications

- **HTTP/TCP/IP metadata** provides valuable evidence for network investigations
- **Packet-level analysis** enables reconstruction of conversations and data transfer
- **Sequence numbers** help identify packet ordering and potential session hijacking
- **Headers** reveal user-agent information, timestamps, and server configuration
- **Content** may contain evidence of unauthorized access or data exfiltration

8.4 Conclusion

This lab successfully demonstrated the fundamental skills required for network forensic analysis of HTTP traffic. The complete session was captured, analyzed, and documented across all protocol layers. Key concepts including TCP connection establishment, HTTP request/response exchange, encapsulation, and evidence preservation were thoroughly examined. The skills acquired in this exercise provide a foundation for more advanced network forensic investigations.

9. APPENDIX

9.1 Complete Commands Used

```
bash
# Setup
mkdir -p ~/SBT-DF203-Lab1/{evidence,working,exported,reports,screenshots,scripts}
sudo apt update && sudo apt install -y apache2 curl wireshark tshark
sudo systemctl enable --now apache2

# Create webpage
printf '<!DOCTYPE html>\n<html><body><h1>SBT-DF203 HTTP Evidence</h1><p>Name: IBRAHI  
M DISEH GARBA</p><p>Reg No: FWSD/11521</p></body></html>\n' | sudo tee /var/www/html/basic.html

# Capture
sudo tshark -i lo -f 'tcp port 80' -w evidence/basic.pcapng
curl --no-keepalive -v http://127.0.0.1/basic.html

# Preserve
cp --preserve=timestamps evidence/basic.pcapng working/basic_working.pcapng
sha256sum evidence/basic.pcapng working/basic_working.pcapng | tee reports/capture_hashes.txt

# Analysis
tshark -r working/basic_working.pcapng -Y 'tcp.stream eq 0 && tcp.flags.syn==1' -T fields -e frame.number -e frame.time -e ip.src -e tcp.srcport -e ip.dst -e tcp.dstport -e tcp.flags -e tcp.seq -e tcp.ack | tee reports/handshake.tsv

tshark -r working/basic_working.pcapng -Y 'http.request' -T fields -e frame.number -e frame.time -e ip.src -e tcp.srcport -e ip.dst -e tcp.dstport -e http.request.method -e http.request.uri -e http.host -e http.user_agent | tee reports/http_requests.tsv

tshark -r working/basic_working.pcapng -Y 'http.response' -T fields -e frame.number -e frame.time -e http.response.code -e http.response.phrase -e http.server -e http.content_type -e http.content_length | tee reports/http_responses.tsv

tshark -r working/basic_working.pcapng -Y 'tcp.flags.fin==1 || tcp.flags.reset==1' -T fields -e frame.number -e frame.time -e ip.src -e tcp.srcport -e ip.dst -e tcp.dstport -e tcp.flags -e tcp.seq -e tcp.ack | tee reports/connection_close.tsv

# Stream Reconstruction
tshark -r working/basic_working.pcapng -q -z follow,tcp,ascii,0 | tee reports/follow_stream.txt
```

9.2 Report Files List

```
text
SBT-DF203-Lab1/
├── evidence/
│   └── basic.pcapng
├── working/
│   └── basic_working.pcapng
├── exported/
├── reports/
│   ├── capture_hashes.txt
│   ├── curl_verbose.txt
│   ├── handshake.tsv
│   ├── http_requests.tsv
│   ├── http_responses.tsv
│   ├── connection_close.tsv
│   └── follow_stream.txt
├── screenshots/
│   ├── screenshot_01_folder_structure.png
│   ├── screenshot_02_apache_status.png
│   ├── screenshot_03_port80_listener.png
│   ├── screenshot_04_webpage_browser.png
│   ├── screenshot_05_curl_verbose.png
│   ├── screenshot_06_capture_running.png
│   ├── screenshot_07_hashes.png
│   ├── screenshot_08_handshake.png
│   ├── screenshot_09_syn_details.png
│   ├── screenshot_10_http_request.png
│   ├── screenshot_11_http_response.png
│   ├── screenshot_12_follow_stream.png
│   ├── screenshot_13_encapsulation.png
│   └── screenshot_14_connection_close.png
└── scripts/
    └── analysis_commands.sh
```

9.3 Academic Integrity Declaration

I confirm that I completed this lab in an authorized environment, preserved the supplied evidence, did not target any third-party system, and accurately documented my own commands, observations, and conclusions.